

K2-30b

R-2292

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_211104 · created 2026-08-17T01:01:46+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459522.9049 +/- 0.0041 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.012
Transit depth (Rp/Rs)^2	2.65 +/- 0.39 [%]
Orbital Inclination (inc)	89.87 +/- 1.13
Airmass coefficient 1 (a1)	1.016 +/- 0.025
Airmass coefficient 2 (a2)	-0.01 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.48 %
Best Comparison Star	#2 - [575, 47]
Optimal Aperture	3.45
Optimal Annulus	11.23
Transit Duration (day)	0.1129 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.012 vs 0.1266 (deviation 2.49 σ)
Duration fit vs published	0.1129 d vs 0.0971 d (deviation 3.42 σ)
Mid-time O-C	16.4 min
Depth SNR	11.2
Residual scatter	2.48 %

Light curve

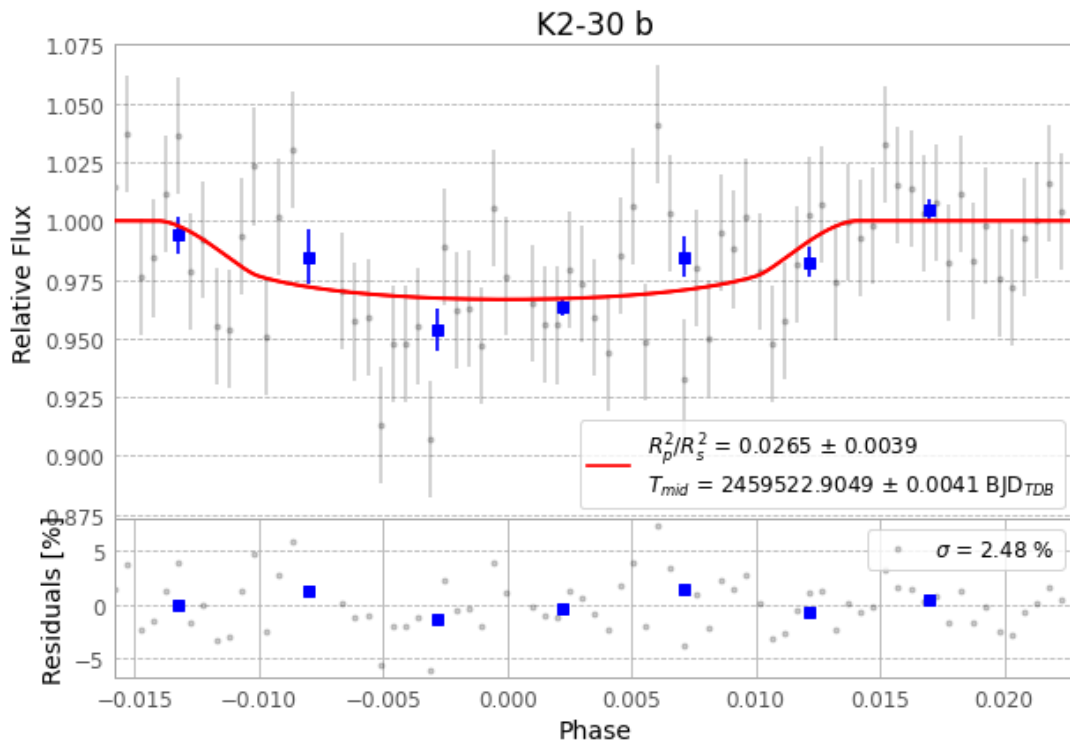


Figure 1 — FinalLightCurve_K2-30 b_2021-11-04.png (SHA-256 e2a508129b7956e7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [472, 243], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 23ef212d8c1b8446...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

74 FITS frames (49 MB), K2-30211104080012.FITS → K2-30211104114812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-211104.log (85a9264e4d6f...), FinalLightCurve_K2-30 b_2021-11-04.png (e2a508129b79...), FinalLightCurve_K2-30 b_2021-11-04.csv (6c7ec23e6a9d...)

Compute resources

Wall time 123.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 01:01Z.